

Integrable Systems

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Most differential equations cannot be solved. This is not a statement about our own cleverness – it is a statement about differential equations, the overwhelming majority of which have solutions that admit no description more economical than the equation itself. Every so often, though, we stumble across an equation that can be ‘integrated up’: one whose solutions can be written down in closed form, or reduced to quadratures, or generated from nothing by an algebraic recipe. Such equations are called *integrable*, and they are the subject of this course.

There is a curious tension here. On the one hand, integrable systems ought to be vanishingly rare, in the same way that a randomly chosen polynomial is unlikely to factorise nicely. On the other hand, integrable equations turn up all over physics – the harmonic oscillator, the Kepler problem, the spinning top, waves in shallow water, light in an optical fibre. Nature seems unreasonably fond of them. The explanation, when we find it, is always the same: integrability is a symptom of hidden structure. An integrable system has more conserved quantities than it has any right to have, and those conserved quantities conspire to fill up phase space with tori on which the motion is nothing more than uniform rotation.

The plan of the course is to take this idea and push it as far as it will go. We begin with ordinary differential equations, where the story is essentially complete: the Arnold–Liouville theorem tells us precisely what it means for a Hamiltonian system to be integrable, and how to integrate it. We then move to partial differential equations, which we can think of as infinite-dimensional dynamical systems. Here the theory is much wilder and much more beautiful. We will meet the Korteweg–de Vries equation and its *solitons* – solitary waves that collide and emerge unscathed, behaving far more like particles than like waves – and we will spend the heart of the course developing the *inverse scattering transform*, an extraordinary machine which solves a nonlinear PDE by turning it into a linear scattering problem, letting the scattering data evolve trivially, and then reconstructing the answer.

This article constitutes my notes for the ‘Integrable Systems’ course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered¹.

Contents

¹We will lean fairly heavily on IB Variational Principles (Hamilton’s equations, the Legendre transform, Noether’s theorem) and on IB Methods (Fourier transforms, Green’s functions, Sturm–Liouville theory). A passing acquaintance with IB Quantum Mechanics – specifically reflection and transmission coefficients for a one-dimensional potential – will make Section 3 feel much more familiar.

1	Integrability of Ordinary Differential Equations	2
1.1	Vector Fields and Flow Maps	2
1.2	Hamiltonian Systems	4
1.3	First Integrals and Liouville Integrability	7
1.4	Canonical Transformations	8
1.4.1	Generating Functions	10
1.5	The Arnold–Liouville Theorem	10
1.6	Action–Angle Variables	16
2	Integrable PDEs and Solitons	19
2.1	Dispersion and Nonlinearity	20
2.2	The Korteweg–de Vries Equation	21
2.3	Travelling Waves and the One-Soliton Solution	23
2.4	Two Solitons, and Why Solitons Are Particles	24
2.5	Conservation Laws	25
2.6	The Miura Transform	27

§1 Integrability of Ordinary Differential Equations

We begin with ordinary differential equations. The eventual goal of this section is the Arnold–Liouville theorem, which says that a Hamiltonian system possessing enough conserved quantities can be solved completely, and moreover tells us how. Before we get there we need to set up some machinery: flows, Poisson brackets, and canonical transformations.

§1.1 Vector Fields and Flow Maps

An m -dimensional first-order ODE is specified by a **vector field** $\mathbf{V} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ together with an initial condition $\mathbf{x}_0 \in \mathbb{R}^m$. We look for a curve $\mathbf{x}(t) \in \mathbb{R}^m$, defined for t in some interval containing 0, satisfying

$$\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x}), \quad \mathbf{x}(0) = \mathbf{x}_0.$$

Restricting to first-order equations costs us nothing: an n th order ODE in one variable becomes a first-order system in n variables in the usual way, by treating the derivatives as new coordinates.

Throughout this course we will be cavalier about regularity, and assume that all our vector fields are as nice as we need them to be.

Fact 1.1

For a sufficiently nice vector field $\mathbf{V}$ and any initial condition $\mathbf{x}_0$, the problem $\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x})$, $\mathbf{x}(0) = \mathbf{x}_0$ has a unique solution, and that solution depends smoothly on both t and $\mathbf{x}_0$.

Because the solution is unique, it is legitimate to package all of the solutions together into a single object.

Definition 1.2 (Flow map)

The **flow map** of a vector field $\mathbf{V}$ is the family of maps $g^t : \mathbb{R}^m \rightarrow \mathbb{R}^m$ defined by

$$\mathbf{x}(t) = g^t \mathbf{x}_0,$$

where $\mathbf{x}(t)$ solves $\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x})$ with $\mathbf{x}(0) = \mathbf{x}_0$.

The flow map is smooth, and satisfies exactly the properties one would hope for.

Proposition 1.3

The flow map obeys

- (i) $g^0 = \text{id}$;
- (ii) $g^{t+s} = g^t g^s$;
- (iii) $(g^t)^{-1} = g^{-t}$.

Proof. The first is immediate from the definition, and the third follows from the first two on setting $s = -t$. For the second, fix $\mathbf{x}_0$ and s , and regard both $t \mapsto g^{t+s} \mathbf{x}_0$ and $t \mapsto g^t(g^s \mathbf{x}_0)$ as functions of t . Both solve

$$\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x}), \quad \mathbf{x}(0) = g^s \mathbf{x}_0,$$

and so by uniqueness of solutions they coincide. $\square$

In the language of IA Groups, this says that $t \mapsto g^t$ is a homomorphism from $(\mathbb{R}, +)$ into the group of diffeomorphisms of $\mathbb{R}^m$. We say that $\mathbf{V}$ is the **infinitesimal generator** of the flow, which is justified by Taylor expanding:

$$g^\varepsilon \mathbf{x}_0 = \mathbf{x}(0) + \varepsilon \dot{\mathbf{x}}(0) + o(\varepsilon) = \mathbf{x}_0 + \varepsilon \mathbf{V}(\mathbf{x}_0) + o(\varepsilon).$$

So to first order in ε , flowing for time ε is just moving along $\mathbf{V}$.

Suppose now that we have two vector fields $\mathbf{V}_1, \mathbf{V}_2$ generating flows g_1^t and g_2^s . A natural question is whether the flows *commute*, that is, whether

$$g_1^t g_2^s \mathbf{x}_0 = g_2^s g_1^t \mathbf{x}_0$$

for all $\mathbf{x}_0, s, t$. Asked directly this is a hopeless question, since it requires us to solve both ODEs. But there is an infinitesimal criterion, and infinitesimal criteria are always easier.

Definition 1.4 (Commutator)

Given vector fields $\mathbf{V}_1, \mathbf{V}_2 : \mathbb{R}^m \rightarrow \mathbb{R}^m$, their **commutator** is the vector field

$$[\mathbf{V}_1, \mathbf{V}_2] = \left(\mathbf{V}_1 \cdot \frac{\partial}{\partial \mathbf{x}} \right) \mathbf{V}_2 - \left(\mathbf{V}_2 \cdot \frac{\partial}{\partial \mathbf{x}} \right) \mathbf{V}_1,$$

where $\frac{\partial}{\partial \mathbf{x}} = (\partial_{x_1}, \dots, \partial_{x_m})^T$. Componentwise,

$$[\mathbf{V}_1, \mathbf{V}_2]_i = \sum_{j=1}^m \left((\mathbf{V}_1)_j \frac{\partial (\mathbf{V}_2)_i}{\partial x_j} - (\mathbf{V}_2)_j \frac{\partial (\mathbf{V}_1)_i}{\partial x_j} \right).$$

Proposition 1.5

Let $\mathbf{V}_1, \mathbf{V}_2$ be vector fields generating flows g_1^t, g_2^s . Then

$$[\mathbf{V}_1, \mathbf{V}_2] = 0 \iff g_1^t g_2^s = g_2^s g_1^t \text{ for all } s, t.$$

We will not prove this (it is a standard exercise, and appears on the first example sheet), but the idea is worth recording: expanding $g_1^{-t} g_2^{-s} g_1^t g_2^s \mathbf{x}_0$ to second order in the small parameters, everything cancels except a term $st [\mathbf{V}_1, \mathbf{V}_2](\mathbf{x}_0)$. The commutator measures precisely the failure of the two flows to commute, and it does so at the level of the vector fields, where we can compute.

§1.2 Hamiltonian Systems

We now specialise to a very particular class of ODEs. In IB Variational Principles we met Hamilton's equations, obtained from a Lagrangian by a Legendre transform; here we take them as our starting point.

The setting is a **phase space** $M = \mathbb{R}^{2n}$, with coordinates

$$(\mathbf{q}, \mathbf{p}) = (q_1, \dots, q_n, p_1, \dots, p_n),$$

which we think of as generalised positions and generalised momenta. We will usually write

$$\mathbf{x} = (\mathbf{q}, \mathbf{p})^T \in M.$$

The crucial point is that M is not just $\mathbb{R}^{2n}$: the coordinates come *paired up*, with q_i married to p_i . This pairing is the extra structure that distinguishes a phase space from a bare vector space, and we encode it in the $2n \times 2n$ antisymmetric matrix

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix},$$

called the **symplectic form**. Note that $J^T = -J$ and $J^2 = -I_{2n}$.

Almost everything in this section can be phrased in terms of J alone. The first example is the Poisson bracket.

Definition 1.6 (Poisson bracket)

For smooth $f, g : M \rightarrow \mathbb{R}$, the **Poisson bracket** is

$$\{f, g\} = \frac{\partial f}{\partial \mathbf{x}} \cdot J \frac{\partial g}{\partial \mathbf{x}} = \frac{\partial f}{\partial \mathbf{q}} \cdot \frac{\partial g}{\partial \mathbf{p}} - \frac{\partial f}{\partial \mathbf{p}} \cdot \frac{\partial g}{\partial \mathbf{q}}.$$

The two expressions agree by direct computation with the block form of J . The Poisson bracket has a list of properties, most of which are obvious and one of which is not.

Proposition 1.7

The Poisson bracket satisfies, for all smooth $f, g, h : M \rightarrow \mathbb{R}$:

- (i) it is bilinear;
- (ii) it is antisymmetric, $\{f, g\} = -\{g, f\}$;
- (iii) it obeys the Leibniz rule $\{f, gh\} = \{f, g\}h + \{f, h\}g$;
- (iv) it obeys the Jacobi identity

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0;$$

- (v) on the coordinate functions,

$$\{q_i, q_j\} = \{p_i, p_j\} = 0, \quad \{q_i, p_j\} = \delta_{ij}.$$

Proof. All of these are immediate from the definition except the Jacobi identity, which one proves by writing out all twenty-four terms and watching them cancel. The only structural remark worth making is that each of the three double brackets contains second derivatives of exactly one of f, g, h , and these second-derivative terms cancel in pairs because J is antisymmetric and second partial derivatives commute. $\square$

The last of these is called the **canonical commutation relation**, and it is the algebraic shadow of the pairing between q_i and p_i .

Definition 1.8 (Hamilton's equations)

Given a function $H : M \rightarrow \mathbb{R}$, called the **Hamiltonian**, **Hamilton's equations** are

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}}. \quad (*)$$

In terms of J these collapse into the single equation

$$\dot{\mathbf{x}} = J \frac{\partial H}{\partial \mathbf{x}}.$$

This is worth pausing over. To specify a general ODE on $\mathbb{R}^{2n}$ we must supply a vector field, which is $2n$ functions. To specify a Hamiltonian system we need only supply the *single* function H . The symplectic structure does the rest of the work. This is our first hint that Hamiltonian systems are special.

Definition 1.9 (Hamiltonian vector field)

Given $f : M \rightarrow \mathbb{R}$, the associated **Hamiltonian vector field** is

$$\mathbf{V}_f = J \frac{\partial f}{\partial \mathbf{x}}.$$

Thus Hamilton's equations say precisely that the motion is the flow of $\mathbf{V}_H$.

Now suppose we are watching a particle move according to $(*)$, and we are interested in some observable $f : M \rightarrow \mathbb{R}$ – its energy, its angular momentum, its distance from the

origin. How does f change along the motion? The chain rule gives

$$\frac{df}{dt} = \frac{d}{dt}f(\mathbf{x}(t)) = \frac{\partial f}{\partial \mathbf{x}} \cdot \dot{\mathbf{x}} = \frac{\partial f}{\partial \mathbf{x}} \cdot J \frac{\partial H}{\partial \mathbf{x}} = \{f, H\}.$$

We record this, because it is the single most useful formula in the subject.

Proposition 1.10

If $\mathbf{x}(t)$ evolves according to Hamilton's equations with Hamiltonian H , then for any smooth $f : M \rightarrow \mathbb{R}$,

$$\frac{df}{dt} = \{f, H\}.$$

In other words – f is conserved if and only if it Poisson commutes with H . This is genuinely powerful. Without it, deciding whether some quantity is conserved means solving the equations of motion and staring at the answer, or else differentiating and hoping that things cancel. With it, we simply compute a bracket. And immediately we get one conserved quantity for free:

$$\frac{dH}{dt} = \{H, H\} = 0$$

by antisymmetry. Energy is conserved.

Example 1.11 (Particle in a potential)

Consider a particle of unit mass with Cartesian position $\mathbf{q} = (q_1, q_2, q_3)$ moving in a potential $U(\mathbf{q})$. Newton's second law reads

$$\ddot{\mathbf{q}} = -\frac{\partial U}{\partial \mathbf{q}}.$$

Introducing $\mathbf{p} = \dot{\mathbf{q}}$, we have

$$\dot{\mathbf{x}} = \begin{pmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{pmatrix} = \begin{pmatrix} \mathbf{p} \\ -\partial U / \partial \mathbf{q} \end{pmatrix} = J \frac{\partial H}{\partial \mathbf{x}}, \quad H = \frac{1}{2}|\mathbf{p}|^2 + U(\mathbf{q}),$$

since $\partial H / \partial \mathbf{p} = \mathbf{p}$ and $\partial H / \partial \mathbf{q} = \partial U / \partial \mathbf{q}$. So Newtonian mechanics in a potential is a Hamiltonian system, and the Hamiltonian is the total energy.

Example 1.12 (Angular momentum in the two-body problem)

Fix the Sun at the origin and let a planet have Cartesian coordinates $\mathbf{q}$. The equation of motion is

$$\ddot{\mathbf{q}} = -\frac{\mathbf{q}}{|\mathbf{q}|^3},$$

which by the previous example is Hamiltonian with $\mathbf{p} = \dot{\mathbf{q}}$ and

$$H = \frac{1}{2}|\mathbf{p}|^2 - \frac{1}{|\mathbf{q}|}.$$

The angular momentum is $\mathbf{L} = \mathbf{q} \wedge \mathbf{p}$, or in components $L_i = \varepsilon_{ijk} q_j p_k$. Let us check

that it is conserved, using summation convention throughout:

$$\begin{aligned}
 \{L_i, H\} &= \frac{\partial L_i}{\partial q_\ell} \frac{\partial H}{\partial p_\ell} - \frac{\partial L_i}{\partial p_\ell} \frac{\partial H}{\partial q_\ell} \\
 &= \varepsilon_{ijk} \delta_{j\ell} p_k \cdot p_\ell - \varepsilon_{ijk} q_j \delta_{k\ell} \cdot \frac{q_\ell}{|\mathbf{q}|^3} \\
 &= \varepsilon_{ijk} \left(p_j p_k - \frac{q_j q_k}{|\mathbf{q}|^3} \right) \\
 &= 0,
 \end{aligned}$$

since in each term we are contracting the antisymmetric ε_{ijk} against something symmetric in $j \leftrightarrow k$. So all three components of $\mathbf{L}$ are conserved, as is H : this system has rather a lot of conserved quantities, which is exactly why the Kepler problem can be solved.

There is one more piece of structure worth recording. We now have two ways of combining functions to produce vector fields: we can take the Poisson bracket $\{f, g\}$ and then form its Hamiltonian vector field, or we can form $\mathbf{V}_f$ and $\mathbf{V}_g$ and take their commutator. These agree up to sign.

Proposition 1.13

For smooth $f, g : M \rightarrow \mathbb{R}$ we have $[\mathbf{V}_f, \mathbf{V}_g] = -\mathbf{V}_{\{f, g\}}$.

The proof is a computation using the Jacobi identity, and is left to the first example sheet. The content of the statement is that $f \mapsto \mathbf{V}_f$ is (up to a sign) a homomorphism of Lie algebras, from functions-under-Poisson-bracket to vector-fields-under-commutator. We will use it in the proof of the Arnold–Liouville theorem, where it will tell us that flows generated by commuting conserved quantities themselves commute.

§1.3 First Integrals and Liouville Integrability

Definition 1.14 (First integral)

Let (M, H) be a Hamiltonian system. A function $f : M \rightarrow \mathbb{R}$ is a **first integral** if

$$\{f, H\} = 0,$$

equivalently, if f is constant along every trajectory.

The terminology is historical. When we solve a differential equation by successive integrations, each integration produces a constant; the constant produced by the first integration was called the first integral. For our purposes, a first integral is simply a constant of the motion.

Our goal for the rest of this section is the following. If a Hamiltonian system has *enough* first integrals, then there is a change of coordinates in which the equations of motion become trivial, and we can integrate them up by inspection. To make ‘enough’ precise we need two conditions.

Definition 1.15 (Involution)

Two first integrals F, G are **in involution** if $\{F, G\} = 0$. We also say that they **Poisson commute**.

Definition 1.16 (Independence)

Functions $f_1, \dots, f_n : M \rightarrow \mathbb{R}$ are **independent** if at each point $\mathbf{x} \in M$ the gradients $\frac{\partial f_i}{\partial \mathbf{x}}$, $i = 1, \dots, n$, are linearly independent vectors in $\mathbb{R}^{2n}$.

Definition 1.17 (Liouville integrable)

A $2n$ -dimensional Hamiltonian system (M, H) is **integrable** (in the Liouville sense) if there exist n first integrals $f_1, \dots, f_n$ which are independent and in involution, that is $\{f_i, f_j\} = 0$ for all i, j .

By convention we always take $f_1 = H$, which is legitimate since $\{H, H\} = 0$.

The independence requirement is what stops the definition being vacuous. Without it we could always cheat, taking $f_1 = H$, $f_2 = H^2$, $f_3 = e^H$, and so on: these all Poisson commute with everything in sight, but they carry no information beyond H itself. Independence says that each new first integral genuinely cuts down the available motion.

The involutivity requirement is subtler and less obviously natural, but it is exactly what makes the proof of the Arnold–Liouville theorem work. It is worth noting that a system can have many first integrals without being integrable in this sense, if the integrals refuse to Poisson commute.

Example 1.18

Every two-dimensional Hamiltonian system ($n = 1$) is integrable: we need one first integral, and H itself will do. This is a restatement of the familiar fact that a single particle moving in one dimension in a potential can always be solved by energy conservation and a quadrature.

Example 1.19

Consider the two-body problem above. We have H , and the three components of $\mathbf{L}$. However

$$\{L_1, L_2\} = L_3 \neq 0,$$

so the L_i are not in involution with each other, and we cannot simply take all four. But H , L_3 and $|\mathbf{L}|^2$ are pairwise in involution and independent, and the system is six-dimensional, so $n = 3$ and the Kepler problem is integrable.

§1.4 Canonical Transformations

We want to find coordinates in which an integrable system becomes trivial. But we cannot use arbitrary coordinates: the whole structure above depended on the pairing of q_i with p_i , and a general change of variables will destroy it. So we must decide which coordinate changes are allowed. There are several equivalent ways to do this; we take the most pragmatic, and simply demand that Hamilton's equations keep their form.

Definition 1.20 (Canonical transformation)

A coordinate change $(\mathbf{q}, \mathbf{p}) \mapsto (\mathbf{Q}(\mathbf{q}, \mathbf{p}), \mathbf{P}(\mathbf{q}, \mathbf{p}))$ is **canonical** if it preserves the form of Hamilton's equations: that is, if

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}} \quad \Longleftrightarrow \quad \dot{\mathbf{Q}} = \frac{\partial \tilde{H}}{\partial \mathbf{P}}, \quad \dot{\mathbf{P}} = -\frac{\partial \tilde{H}}{\partial \mathbf{Q}},$$

where $\tilde{H}(\mathbf{Q}, \mathbf{P}) = H(\mathbf{q}, \mathbf{p})$.

Writing $\mathbf{x} = (\mathbf{q}, \mathbf{p})$ and $\mathbf{y} = (\mathbf{Q}, \mathbf{P})$, this says

$$\dot{\mathbf{x}} = J \frac{\partial H}{\partial \mathbf{x}} \Longleftrightarrow \dot{\mathbf{y}} = J \frac{\partial \tilde{H}}{\partial \mathbf{y}}.$$

Example 1.21

Swapping $\mathbf{q}$ and $\mathbf{p}$ is *not* canonical: it introduces a sign error, turning $\dot{\mathbf{Q}} = \partial \tilde{H} / \partial \mathbf{P}$ into $\dot{\mathbf{Q}} = -\partial \tilde{H} / \partial \mathbf{P}$. The map $(\mathbf{q}, \mathbf{p}) \mapsto (\mathbf{p}, -\mathbf{q})$, on the other hand, is canonical.

Let us do the linear case honestly, since it tells us what the general answer must look like.

Example 1.22 (Linear canonical transformations)

Consider $\mathbf{x} \mapsto \mathbf{y} = A\mathbf{x}$ for a constant matrix A . We claim this is canonical if and only if $AJA^T = J$, in which case A is called **symplectic**.

Indeed, by linearity $\dot{\mathbf{y}} = A\dot{\mathbf{x}} = AJ \frac{\partial H}{\partial \mathbf{x}}$. Setting $\tilde{H}(\mathbf{y}) = H(\mathbf{x})$ and using the chain rule,

$$\frac{\partial H}{\partial x_i} = \frac{\partial y_j}{\partial x_i} \frac{\partial \tilde{H}}{\partial y_j} = A_{ji} \frac{\partial \tilde{H}}{\partial y_j} = \left[A^T \frac{\partial \tilde{H}}{\partial \mathbf{y}} \right]_i.$$

Substituting back,

$$\dot{\mathbf{y}} = AJA^T \frac{\partial \tilde{H}}{\partial \mathbf{y}},$$

which is Hamilton's equation for $\tilde{H}$ precisely when $AJA^T = J$.

Since a differentiable map is locally linear, and Hamilton's equations are a local condition, the general statement is the obvious one.

Proposition 1.23

A smooth map $\mathbf{x} \mapsto \mathbf{y}(\mathbf{x})$ is canonical if and only if its Jacobian $D\mathbf{y}$ is symplectic at every point:

$$D\mathbf{y} J (D\mathbf{y})^T = J.$$

This is a clean criterion, but it is a criterion for *checking* that a transformation is canonical, not for *producing* one. For that we use generating functions.

§1.4.1 Generating Functions

There are four standard flavours of generating function, differing in which pair of old and new variables one takes as independent. We will need only one of them.

Let $S : \mathbb{R}^{2n} \rightarrow \mathbb{R}$ be a function which we write as $S(\mathbf{q}, \mathbf{P})$ – old positions and new momenta – and define a transformation implicitly by

$$\mathbf{p} = \frac{\partial S}{\partial \mathbf{q}}, \quad \mathbf{Q} = \frac{\partial S}{\partial \mathbf{P}}.$$

These equations are to be read as follows. The first is n equations relating $\mathbf{p}, \mathbf{q}, \mathbf{P}$; we solve them for $\mathbf{P}$ in terms of $(\mathbf{q}, \mathbf{p})$. The second then delivers $\mathbf{Q}$ in terms of $(\mathbf{q}, \mathbf{P})$, hence in terms of $(\mathbf{q}, \mathbf{p})$. Checking that the result is always canonical is another careful application of the chain rule, which we omit.

In practice we usually already have a candidate for $\mathbf{P}$ in mind – typically some collection of first integrals – and we hunt for an S making the first equation hold. The second equation then *tells* us what the conjugate variable $\mathbf{Q}$ has to be. This is exactly how the Arnold–Liouville theorem will proceed.

Example 1.24

Take $S(\mathbf{q}, \mathbf{P}) = \mathbf{q} \cdot \mathbf{P}$. Then

$$\mathbf{p} = \frac{\partial S}{\partial \mathbf{q}} = \mathbf{P}, \quad \mathbf{Q} = \frac{\partial S}{\partial \mathbf{P}} = \mathbf{q},$$

so S generates the identity transformation.

Example 1.25

In two dimensions, take $S(q, P) = qP + q^2$. Then

$$p = \frac{\partial S}{\partial q} = P + 2q, \quad Q = \frac{\partial S}{\partial P} = q,$$

so the transformation is $(Q, P) = (q, p - 2q)$, or in matrix form

$$\begin{pmatrix} Q \\ P \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} q \\ p \end{pmatrix} = A \begin{pmatrix} q \\ p \end{pmatrix}.$$

We can verify directly that this is canonical:

$$AJA^T = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = J,$$

as promised.

§1.5 The Arnold–Liouville Theorem

We can now state the central result about integrable ODEs. Informally it says: if a Hamiltonian system is integrable, then there is a canonical transformation $(\mathbf{q}, \mathbf{p}) \mapsto (\phi, \mathbf{I})$ after which the new Hamiltonian depends only on $\mathbf{I}$. If we can achieve this then

Hamilton's equations become

$$\dot{\mathbf{I}} = -\frac{\partial \tilde{H}}{\partial \phi} = 0, \quad \dot{\phi} = \frac{\partial \tilde{H}}{\partial \mathbf{I}},$$

so $\mathbf{I}(t) = \mathbf{I}_0$ is constant, and since the right-hand side of the second equation depends on $\mathbf{I}$ alone, $\dot{\phi}$ is constant too. Hence

$$\phi(t) = \phi_0 + \Omega t, \quad \Omega = \frac{\partial \tilde{H}}{\partial \mathbf{I}}(\mathbf{I}_0),$$

and we have solved the system completely. All the difficulty has been pushed into finding the transformation.

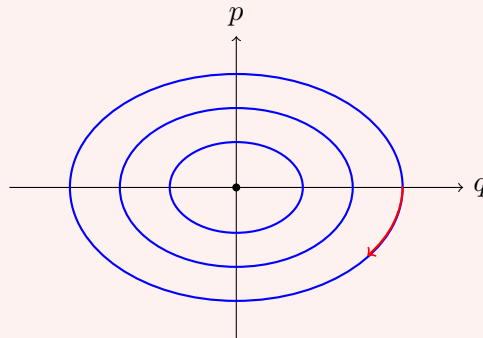
Before proving anything, let us see what the transformation looks like in the simplest possible example, where we can produce it by divine inspiration and then reverse-engineer where it came from.

Example 1.26 (Harmonic oscillator, first pass)

Take the harmonic oscillator on $M = \mathbb{R}^2$, with

$$H(q, p) = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2.$$

This is a two-dimensional system, so it is automatically integrable, with the single first integral $f_1 = H$. The level sets of H are ellipses:



Each of these is homeomorphic to a circle S^1 , and this is not an accident – it is the first half of the Arnold–Liouville theorem in miniature.

Now introduce new coordinates (ϕ, I) by

$$q = \sqrt{\frac{2I}{\omega}} \sin \phi, \quad p = \sqrt{2I\omega} \cos \phi.$$

One can check by computing the Jacobian that this is canonical. In these coordinates,

$$\tilde{H}(\phi, I) = \frac{1}{2} \cdot 2I\omega \cos^2 \phi + \frac{1}{2}\omega^2 \cdot \frac{2I}{\omega} \sin^2 \phi = \omega I.$$

There is no ϕ ! So Hamilton's equations read

$$\dot{\phi} = \frac{\partial \tilde{H}}{\partial I} = \omega, \quad \dot{I} = -\frac{\partial \tilde{H}}{\partial \phi} = 0,$$

giving $\phi(t) = \phi_0 + \omega t$ and $I(t) = I_0$. The oscillator goes round its ellipse at constant angular rate ω , which is of course the right answer.

Where could that transformation have come from? Here is a clue. Let us compute, for no obvious reason, the area enclosed by a level set of H , divided by 2π :

$$\begin{aligned} \frac{1}{2\pi} \oint p \, dq &= \frac{1}{2\pi} \int_0^{2\pi} p(\phi, I) \left(\frac{\partial q}{\partial \phi} d\phi + \frac{\partial q}{\partial I} dI \right) \\ &= \frac{1}{2\pi} \int_0^{2\pi} p(\phi, I) \frac{\partial q}{\partial \phi} d\phi \\ &= \frac{1}{2\pi} \int_0^{2\pi} \sqrt{2I\omega} \cos \phi \cdot \sqrt{\frac{2I}{\omega}} \cos \phi \, d\phi \\ &= \frac{2I}{2\pi} \int_0^{2\pi} \cos^2 \phi \, d\phi = I, \end{aligned}$$

using $dI = 0$ along a level set. So the new momentum I is nothing but the enclosed area over 2π – a quantity we could have computed knowing nothing whatsoever about the transformation.

There are two lessons here, and they are exactly the two halves of the theorem: the motion takes place on a circle, and the right momentum coordinate is $\frac{1}{2\pi} \oint p \, dq$.

Theorem 1.27 (Arnold–Liouville)

Let (M, H) be an integrable $2n$ -dimensional Hamiltonian system with independent, involutive first integrals $f_1, \dots, f_n$, where $f_1 = H$. For $\mathbf{c} \in \mathbb{R}^n$ set

$$M_{\mathbf{c}} = \{(\mathbf{q}, \mathbf{p}) \in M : f_i(\mathbf{q}, \mathbf{p}) = c_i, \, i = 1, \dots, n\}.$$

Then:

- (i) $M_{\mathbf{c}}$ is a smooth n -dimensional surface in M , invariant under the flow. If $M_{\mathbf{c}}$ is compact and connected, it is diffeomorphic to the n -torus

$$T^n = \underbrace{S^1 \times \dots \times S^1}_n.$$

- (ii) If $M_{\mathbf{c}}$ is compact and connected, then locally there is a canonical transformation $(\mathbf{q}, \mathbf{p}) \mapsto (\phi, \mathbf{I})$ to **action–angle coordinates**, in which the ϕ_k are angular coordinates on $M_{\mathbf{c}}$, the actions I_k are first integrals, and $\tilde{H}$ is independent of ϕ . Consequently

$$\dot{\mathbf{I}} = 0, \quad \dot{\phi} = \frac{\partial \tilde{H}}{\partial \mathbf{I}} = \text{constant}.$$

The first part is pure differential geometry, and the second is where the work is. We give the argument in outline; readers who have not met the preimage theorem or the orbit–stabiliser theorem in this setting may take the geometric steps on faith.

Proof sketch. Part (i). That $M_{\mathbf{c}}$ is a smooth n -dimensional surface is the preimage theorem (a consequence of the inverse function theorem from IB Analysis and

Topology) applied to $\mathbf{f} = (f_1, \dots, f_n) : M \rightarrow \mathbb{R}^n$. What makes it apply is precisely the independence of the f_i , which says that $\mathbf{c}$ is a regular value.

Next we show that $M_{\mathbf{c}}$ is a torus when compact and connected. Consider the Hamiltonian vector fields

$$\mathbf{V}_{f_i} = J \frac{\partial f_i}{\partial \mathbf{x}}.$$

We claim each is tangent to $M_{\mathbf{c}}$. It suffices to show that the derivative of each f_j along $\mathbf{V}_{f_i}$ vanishes, and indeed

$$\left(\mathbf{V}_{f_i} \cdot \frac{\partial}{\partial \mathbf{x}} \right) f_j = \frac{\partial f_j}{\partial \mathbf{x}} \cdot J \frac{\partial f_i}{\partial \mathbf{x}} = \{f_j, f_i\} = 0$$

by involutivity. So the flows g_i^t generated by the $\mathbf{V}_{f_i}$ map $M_{\mathbf{c}}$ into itself. Moreover these flows commute, since

$$[\mathbf{V}_{f_i}, \mathbf{V}_{f_j}] = -\mathbf{V}_{\{f_i, f_j\}} = -\mathbf{V}_0 = 0.$$

So writing $\mathbf{t} = (t_1, \dots, t_n) \in \mathbb{R}^n$ and

$$g^{\mathbf{t}} = g_1^{t_1} g_2^{t_2} \cdots g_n^{t_n},$$

commutativity gives $g^{\mathbf{t}_1 + \mathbf{t}_2} = g^{\mathbf{t}_1} g^{\mathbf{t}_2}$, so we have an action of the group $(\mathbb{R}^n, +)$ on $M_{\mathbf{c}}$.

Fix $\mathbf{x} \in M_{\mathbf{c}}$ and let

$$\text{Stab}(\mathbf{x}) = \{\mathbf{t} \in \mathbb{R}^n : g^{\mathbf{t}} \mathbf{x} = \mathbf{x}\}.$$

By the orbit–stabiliser theorem, $\mathbf{t} \mapsto g^{\mathbf{t}} \mathbf{x}$ descends to a bijection $\mathbb{R}^n / \text{Stab}(\mathbf{x}) \rightarrow \text{Orb}(\mathbf{x})$, and one checks that the orbit of $\mathbf{x}$ is the connected component of $\mathbf{x}$ in $M_{\mathbf{c}}$ – so if $M_{\mathbf{c}}$ is connected, the orbit is everything. Now $\text{Stab}(\mathbf{x})$ is a subgroup of $\mathbb{R}^n$, and (provided the flows are non-degenerate) it is discrete, hence isomorphic to $\mathbb{Z}^k$ for some $0 \leq k \leq n$. Therefore

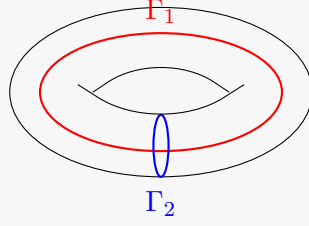
$$M_{\mathbf{c}} \cong \mathbb{R}^n / \mathbb{Z}^k \cong T^k \times \mathbb{R}^{n-k}.$$

If $M_{\mathbf{c}}$ is compact there can be no $\mathbb{R}$ factors, so $k = n$ and $M_{\mathbf{c}} \cong T^n$.

Part (ii). We construct the action–angle coordinates. For clarity of presentation we take $n = 2$; the general case is identical except that we need a higher-dimensional version of Green’s theorem, which we do not have to hand.

It is trivially easy to find *some* coordinates $(\mathbf{Q}, \mathbf{P})$ with $\mathbf{P}$ constant along the flow and $\mathbf{Q}$ angular: just set $\mathbf{P} = \mathbf{c}$ and use the diffeomorphism $M_{\mathbf{c}} \cong T^n$ to get angles. The problem is that such a transformation will almost never be canonical. So we must be cleverer about the choice of $\mathbf{P}$.

On the torus T^n there are n topologically distinct ‘basic’ loops $\Gamma_1, \dots, \Gamma_n$ – under the identification $M_{\mathbf{c}} \cong S^1 \times \cdots \times S^1$, the loop Γ_j runs once around the j th factor and is constant on the others. For $n = 2$ these are the two obvious circles:



We now define the **actions**

$$I_j = \frac{1}{2\pi} \oint_{\Gamma_j} \mathbf{p} \cdot d\mathbf{q},$$

generalising the formula we found for the harmonic oscillator. The first thing to check is that this is well defined, since Γ_j is really a whole homotopy class of loops rather than a single one. So suppose Γ_2 and Γ'_2 are two such loops, bounding between them an annular region A on the torus.

On $M_{\mathbf{c}}$ the equations $f_i(\mathbf{q}, \mathbf{p}) = c_i$ hold identically, and we assume we may invert them locally to write $\mathbf{p} = \mathbf{p}(\mathbf{q}, \mathbf{c})$; the condition for this is $\det(\partial f_i / \partial p_j) \neq 0$. Differentiating the identity $f_i(\mathbf{q}, \mathbf{p}(\mathbf{q}, \mathbf{c})) = c_i$ with respect to q_k gives

$$\frac{\partial f_i}{\partial q_k} + \frac{\partial f_i}{\partial p_\ell} \frac{\partial p_\ell}{\partial q_k} = 0$$

on $M_{\mathbf{c}}$. Now use involutivity: on $M_{\mathbf{c}}$,

$$\begin{aligned} 0 &= \{f_i, f_j\} \\ &= \frac{\partial f_i}{\partial q_k} \frac{\partial f_j}{\partial p_k} - \frac{\partial f_i}{\partial p_k} \frac{\partial f_j}{\partial q_k} \\ &= \left(-\frac{\partial f_i}{\partial p_\ell} \frac{\partial p_\ell}{\partial q_k} \right) \frac{\partial f_j}{\partial p_k} - \frac{\partial f_i}{\partial p_k} \left(-\frac{\partial f_j}{\partial p_\ell} \frac{\partial p_\ell}{\partial q_k} \right) \\ &= \frac{\partial f_i}{\partial p_k} \left(\frac{\partial p_\ell}{\partial q_k} - \frac{\partial p_k}{\partial q_\ell} \right) \frac{\partial f_j}{\partial p_\ell}, \end{aligned}$$

where in the last line we relabelled dummy indices in the first term. Since the matrices $\partial f_i / \partial p_k$ are invertible, the matrix in the middle must vanish:

$$\frac{\partial p_\ell}{\partial q_k} = \frac{\partial p_k}{\partial q_\ell}.$$

For $n = 2$ the only content of this is $\partial p_1 / \partial q_2 = \partial p_2 / \partial q_1$. But then by Green's theorem

$$\left(\oint_{\Gamma_2} - \oint_{\Gamma'_2} \right) \mathbf{p} \cdot d\mathbf{q} = \oint_{\partial A} \mathbf{p} \cdot d\mathbf{q} = \iint_A \left(\frac{\partial p_2}{\partial q_1} - \frac{\partial p_1}{\partial q_2} \right) dq_1 dq_2 = 0.$$

So I_j depends only on the homotopy class of Γ_j , and hence $\mathbf{I} = \mathbf{I}(\mathbf{c})$ is a function of $\mathbf{c}$ alone. These are our new momenta. We assume $\mathbf{I}(\mathbf{c})$ is invertible, so that we may also write $\mathbf{c} = \mathbf{c}(\mathbf{I})$.

It remains to produce the conjugate angles, and for this we use a generating function. Pick a base point $\mathbf{x}_0 \in M_{\mathbf{c}}$ and set

$$S(\mathbf{q}, \mathbf{I}) = \int_{\mathbf{x}_0}^{\mathbf{x}} \mathbf{p}(\mathbf{q}', \mathbf{c}(\mathbf{I})) \cdot d\mathbf{q}',$$

where $\mathbf{x} = (\mathbf{q}, \mathbf{p}(\mathbf{q}, \mathbf{c}(\mathbf{I})))$ and the integral is along any path in $M_{\mathbf{c}}$. Is this well defined? If we deform the path slightly, sweeping out a region B , then S changes by $\oint_{\partial B} \mathbf{p} \cdot d\mathbf{q}$, which vanishes by exactly the Green's theorem computation above. But we are on a torus, and there are paths that cannot be deformed into one another: a path may wind around one of the cycles Γ_j on its way. Adding such a loop changes S by $\oint_{\Gamma_j} \mathbf{p} \cdot d\mathbf{q} = 2\pi I_j$. So the total ambiguity is

$$S(\mathbf{q}, \mathbf{I}) \mapsto S(\mathbf{q}, \mathbf{I}) + 2\pi m_j I_j, \quad m_j \in \mathbb{Z},$$

and nothing worse. Differentiating with respect to $\mathbf{I}$, we conclude that

$$\phi = \frac{\partial S}{\partial \mathbf{I}}$$

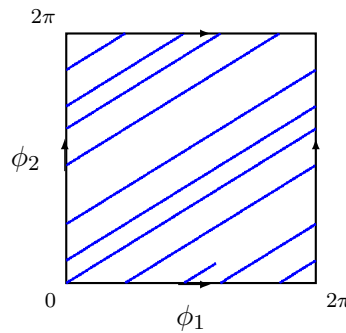
is well defined *modulo* 2π . These are the angles, and the ambiguity is exactly the ambiguity one expects of an angular coordinate.

Finally we should check that S really is a generating function in the sense of the previous subsection, i.e. that $\partial S / \partial \mathbf{q} = \mathbf{p}$. Writing $S = \int_{\mathbf{x}_0}^{\mathbf{x}} \mathbf{F} \cdot d\mathbf{x}'$ with $\mathbf{F} = (\mathbf{p}, \mathbf{0})$, the fundamental theorem of calculus gives $\partial S / \partial \mathbf{x} = \mathbf{F}$, and in particular $\partial S / \partial \mathbf{q} = \mathbf{p}$ as required.

So S generates a canonical transformation $(\mathbf{q}, \mathbf{p}) \mapsto (\phi, \mathbf{I})$. Since $\mathbf{I} = \mathbf{I}(\mathbf{c}) = \mathbf{I}(\mathbf{f}(\mathbf{x}))$ is a function of the first integrals alone, we have $\dot{\mathbf{I}} = 0$; and since $\dot{\mathbf{I}} = -\partial \tilde{H} / \partial \phi$, the new Hamiltonian is independent of ϕ . Integrating up,

$$\phi(t) = \phi_0 + \Omega t, \quad \mathbf{I}(t) = \mathbf{I}_0, \quad \Omega = \frac{\partial \tilde{H}}{\partial \mathbf{I}}(\mathbf{I}_0). \quad \square$$

The picture that emerges is a very appealing one. Phase space is foliated by invariant tori $M_{\mathbf{c}}$, one for each value of the conserved quantities, and on each torus the motion is uniform straight-line motion in the angle variables – a winding line, going round the j th cycle at frequency Ω_j . Drawing the torus in the usual way as a square with opposite edges identified, the trajectory is simply a straight line of slope Ω_2 / Ω_1 :



If the frequencies $\Omega_1, \dots, \Omega_n$ are rationally related the trajectory closes up and the motion is periodic; if they are rationally independent the trajectory winds around forever,

densely filling the torus, and the motion is called **quasi-periodic**.

§1.6 Action–Angle Variables

The proof above is constructive, and gives us a recipe. To integrate up an integrable Hamiltonian system:

- (1) Identify the n cycles $\Gamma_1, \dots, \Gamma_n$ on the invariant torus $M_{\mathbf{c}}$, and solve $f_i(\mathbf{q}, \mathbf{p}) = c_i$ for $\mathbf{p} = \mathbf{p}(\mathbf{q}, \mathbf{c})$.
- (2) Compute the actions

$$I_j = \frac{1}{2\pi} \oint_{\Gamma_j} \mathbf{p} \cdot d\mathbf{q},$$

and invert to obtain $\mathbf{c} = \mathbf{c}(\mathbf{I})$.

- (3) Form the generating function

$$S(\mathbf{q}, \mathbf{I}) = \int^{\mathbf{q}} \mathbf{p}(\mathbf{q}', \mathbf{c}(\mathbf{I})) \cdot d\mathbf{q}'$$

and read off the angles $\phi = \partial S / \partial \mathbf{I}$.

- (4) Write $\tilde{H} = \tilde{H}(\mathbf{I})$ and read off the frequencies $\boldsymbol{\Omega} = \partial \tilde{H} / \partial \mathbf{I}$.

In practice step (4) is often all one wants: the frequencies of an integrable system can be extracted without ever computing the angles, since $\tilde{H}(\mathbf{I})$ is determined by step (2) alone.

Let us run the recipe on the harmonic oscillator, and see the transformation appear without any divine intervention.

Example 1.28 (Harmonic oscillator, second pass)

With $H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2$ we have

$$M_c = \{(q, p) : \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2 = c\},$$

an ellipse, which is indeed diffeomorphic to $T^1 = S^1$ as the theorem promises. There is only one cycle, the ellipse itself. Solving for p ,

$$p = p(q, c) = \pm \sqrt{2c - \omega^2 q^2}.$$

The action is the enclosed area over 2π . The ellipse $\frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2 = c$ has semi-axes $\sqrt{2c}/\omega$ and $\sqrt{2c}$, so its area is $\pi \cdot \frac{\sqrt{2c}}{\omega} \cdot \sqrt{2c} = 2\pi c/\omega$ and

$$I = \frac{1}{2\pi} \oint p \, dq = \frac{c}{\omega}.$$

Inverting, $c = c(I) = \omega I$, so immediately

$$\tilde{H} = c = \omega I, \quad \Omega = \frac{d\tilde{H}}{dI} = \omega.$$

The frequency is ω , independent of amplitude – the defining feature of the harmonic oscillator.

For the angle, take the base point to be $q = 0$ with $p > 0$. Then

$$S(q, I) = \int_0^q \sqrt{2\omega I - \omega^2 q'^2} \, dq',$$

and differentiating under the integral sign,

$$\phi = \frac{\partial S}{\partial I} = \int_0^q \frac{\omega \, dq'}{\sqrt{2\omega I - \omega^2 q'^2}} = \int_0^q \frac{dq'}{\sqrt{2I/\omega - q'^2}} = \sin^{-1} \left(\sqrt{\frac{\omega}{2I}} q \right).$$

Sure enough, this is well defined only up to multiples of 2π . Inverting,

$$q = \sqrt{\frac{2I}{\omega}} \sin \phi, \quad p = \frac{\partial S}{\partial q} = \sqrt{2\omega I - \omega^2 q^2} = \sqrt{2I\omega} \cos \phi,$$

which is exactly the transformation we pulled out of thin air earlier.

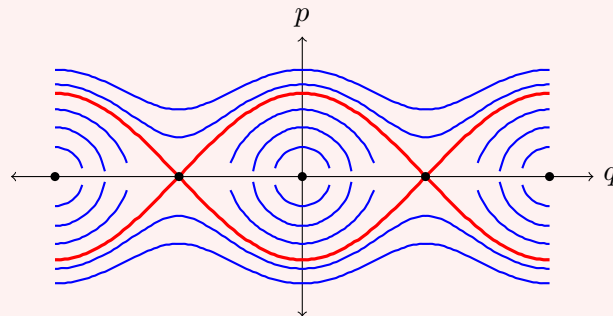
The next example is the one to remember, because it is the simplest system in which the frequency actually depends on the action.

Example 1.29 (The simple pendulum)

A pendulum of unit mass and unit length in unit gravity has Hamiltonian

$$H(q, p) = \frac{1}{2}p^2 - \cos q,$$

where q is the angle from the downward vertical. The phase portrait is the following, with the equilibria marked: centres at $q = 0, \pm 2\pi, \dots$ (the pendulum hanging down) and saddles at $q = \pm\pi$ (the pendulum balanced upright).



There are three qualitatively different kinds of motion, separated by the thick red curve.

- (i) For $-1 < E < 1$ the level set $\{H = E\}$ is a closed curve encircling the origin. This is **libration**: the pendulum swings back and forth about the bottom, with q bounded by the turning points $\pm q_0$ where $\cos q_0 = -E$.
- (ii) For $E > 1$ the level set consists of two curves, one with $p > 0$ throughout and one with $p < 0$. This is **rotation**: the pendulum has enough energy to go over the top, and q increases (or decreases) without bound.
- (iii) At $E = 1$ we have the **separatrix**, $p = \pm 2 \cos(q/2)$, on which the pendulum takes infinite time to creep up to the inverted position $q = \pi$.

Note that the phase space here is really a cylinder rather than a plane, since q

and $q + 2\pi$ describe the same configuration; on the cylinder the libration curves are circles, and so are the rotation curves, but the two families of circles are not homotopic to one another.

Consider the librating region. Solving for p ,

$$p = \pm \sqrt{2(E + \cos q)},$$

and the action is

$$I(E) = \frac{1}{2\pi} \oint p \, dq = \frac{1}{\pi} \int_{-q_0}^{q_0} \sqrt{2(E + \cos q)} \, dq,$$

where $q_0 = \cos^{-1}(-E)$ and the factor of two from the upper and lower halves of the curve has been absorbed. Using $\cos q = 1 - 2 \sin^2(q/2)$ and writing $k^2 = (1 + E)/2 \in (0, 1)$, this becomes

$$I(E) = \frac{4}{\pi} \int_0^{q_0} \sqrt{k^2 - \sin^2(q/2)} \, dq = \frac{8}{\pi} [\mathcal{E}(k) - (1 - k^2)\mathcal{K}(k)],$$

where $\mathcal{K}$ and $\mathcal{E}$ are the complete elliptic integrals of the first and second kind. This is not an elementary function, and that is the point: the pendulum is integrable, but its action–angle variables involve elliptic functions. The frequency is

$$\Omega(I) = \frac{dE}{dI} = \left(\frac{dI}{dE} \right)^{-1} = \frac{\pi}{2\mathcal{K}(k)},$$

using $dI/dE = \frac{2}{\pi}\mathcal{K}(k)$, which follows from differentiating the integral for I under the integral sign.

It is worth sanity-checking the two limits. As $E \rightarrow -1^+$ we have $k \rightarrow 0$, and $\mathcal{K}(0) = \pi/2$, so $\Omega \rightarrow 1$: small oscillations have unit frequency, in agreement with linearising $\ddot{q} = -\sin q \approx -q$. As $E \rightarrow 1^-$ we have $k \rightarrow 1$ and $\mathcal{K}(k) \rightarrow \infty$, so $\Omega \rightarrow 0$: the period diverges as we approach the separatrix, which is the statement that the pendulum takes forever to reach the top. The pendulum is thus the standard example of an integrable system whose frequency depends on its amplitude.

Our final example has two degrees of freedom, so that we get an honest two-torus and two genuinely different cycles.

Example 1.30 (A particle in a central potential)

Consider a particle of unit mass moving in a plane under a central potential $U(r)$. Using polar coordinates (r, θ) , the Hamiltonian is

$$H = \frac{1}{2} \left(p_r^2 + \frac{p_\theta^2}{r^2} \right) + U(r),$$

with the canonical pairs (r, p_r) and (θ, p_θ) . This is a four-dimensional phase space, so we need two first integrals in involution, and we have them: $f_1 = H$ and $f_2 = p_\theta$, the angular momentum. They are in involution because H does not depend on θ :

$$\{p_\theta, H\} = -\frac{\partial H}{\partial \theta} = 0.$$

So the system is integrable. Fix $\mathbf{c} = (E, L)$. The invariant surface $M_{\mathbf{c}}$ is described by

$$p_{\theta} = L, \quad p_r = \pm \sqrt{2(E - U(r)) - \frac{L^2}{r^2}},$$

and provided the effective potential $U_{\text{eff}}(r) = U(r) + L^2/2r^2$ has a well, so that r oscillates between turning points r_- and r_+ , this really is a torus: θ runs around a circle, and (r, p_r) traces a closed curve in its own plane.

The two cycles are then evident. Taking Γ_1 to be the loop on which θ increases by 2π at fixed r , and Γ_2 the loop on which (r, p_r) traverses its closed curve at fixed θ , we get

$$I_1 = \frac{1}{2\pi} \oint_{\Gamma_1} p_{\theta} d\theta = L, \quad I_2 = \frac{1}{2\pi} \oint_{\Gamma_2} p_r dr = \frac{1}{\pi} \int_{r_-}^{r_+} \sqrt{2(E - U(r)) - \frac{L^2}{r^2}} dr.$$

The first action is just the angular momentum, as one would hope. The second is the **radial action**, and computing it gives E as a function of (I_1, I_2) and hence both frequencies at once.

For the Kepler potential $U(r) = -1/r$ this integral can be done by contour methods, with the answer

$$I_2 = -I_1 + \frac{1}{\sqrt{-2E}}, \quad \text{i.e.} \quad \tilde{H} = E = -\frac{1}{2(I_1 + I_2)^2}.$$

The remarkable thing here is that $\tilde{H}$ depends on I_1 and I_2 only through the combination $I_1 + I_2$, so that

$$\Omega_1 = \frac{\partial \tilde{H}}{\partial I_1} = \frac{\partial \tilde{H}}{\partial I_2} = \Omega_2 = \frac{1}{(I_1 + I_2)^3}.$$

The two frequencies are equal, so the winding line on the torus closes up after one circuit: Kepler orbits are closed curves, which is the statement that planetary orbits are ellipses that do not precess. This **degeneracy** is special to the $1/r$ potential (and to the harmonic oscillator), and its breaking by relativistic corrections is what produces the precession of the perihelion of Mercury.

This is a good point to pause and take stock. We have a complete and satisfying theory: a Hamiltonian system with enough commuting conserved quantities lives on invariant tori, and on those tori the motion is trivial. For the rest of the course we ask whether anything like this survives when we pass to partial differential equations. The answer, remarkably, is yes – but almost every ingredient will have to be reinvented.

§2 Integrable PDEs and Solitons

For the rest of the course we turn to partial differential equations. It is helpful to think of a PDE as an infinite-dimensional dynamical system: instead of finitely many coordinates $x_i(t)$ we have a whole function $u(x, t)$, with the spatial variable x playing the role of the index i .

What should we expect of an integrable PDE? If a $2n$ -dimensional ODE is integrable when it has n first integrals, and our phase space is now infinite-dimensional, then – since

half of infinity is still infinity – we should expect an integrable PDE to possess *infinitely many* conserved quantities. We should also expect some magic transformation, analogous to the passage to action–angle variables, which turns the equation into something trivial. Both of these expectations turn out to be correct. What is much less obvious is how to find the transformation, and we will spend the whole of Section 3 constructing it for one particular equation.

We will not attempt to define integrability for PDEs. Nobody has managed to do so in a way that is both precise and useful, and we will be content to work with examples.

§2.1 Dispersion and Nonlinearity

The equation at the centre of this course is the Korteweg–de Vries equation

$$u_t + u_{xxx} - 6uu_x = 0.$$

Before meeting it properly, it pays to understand the two terms u_{xxx} and $-6uu_x$ separately, because each on its own does something violent to a wave, and the whole point of KdV is that the two violences cancel.

Example 2.1 (Dispersion)

Consider the *linear* equation

$$u_t + u_{xxx} = 0.$$

Being linear and constant-coefficient, it is solved by Fourier methods (IB Methods). Plane wave modes $u = e^{i(kx - \omega t)}$ solve it provided ω satisfies the **dispersion relation**

$$\omega = \omega(k) = -k^3.$$

The general solution is then a superposition,

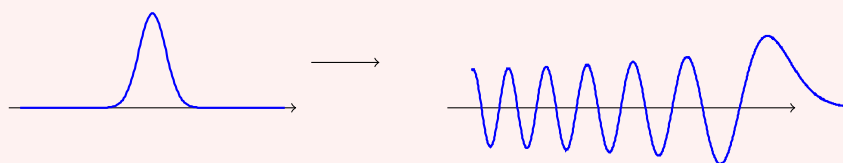
$$u(x, t) = \int_{-\infty}^{\infty} A(k) e^{i(kx - \omega(k)t)} dk,$$

with $A(k) = \hat{u}_0(k)/2\pi$ determined by the initial data.

Writing a single mode as

$$u(x, t) = \exp \left(ik \left(x - \frac{\omega(k)}{k} t \right) \right),$$

we see that it travels with phase speed $\omega/k = -k^2$. The crucial point is that this depends on k : modes of different wavelength travel at different speeds. So an initial profile, which is a superposition of many modes, does not translate rigidly – it spreads out, with the short wavelengths racing off in one direction and the long wavelengths lagging behind. This is **dispersion**, and it is the work of the u_{xxx} term. A localised initial bump decays into an oscillatory wave train:



Had we written $u_t + u_x = 0$ instead, the dispersion relation would be $\omega = k$, the phase speed would be 1 for every mode, and the initial profile would translate rigidly. It is the *third* derivative that does the damage.

Example 2.2 (Wave breaking)

Now consider the *nonlinear* equation

$$u_t - 6uu_x = 0.$$

This looks worse, being nonlinear, but in fact it can be solved exactly for any initial data $u(x, 0) = f(x)$ by the method of characteristics. The characteristics are the curves along which u is constant, and since the equation says u is transported at speed $-6u$, they are straight lines. The solution is given implicitly by

$$u(x, t) = f(\xi), \quad \xi = x - 6tf(\xi).$$

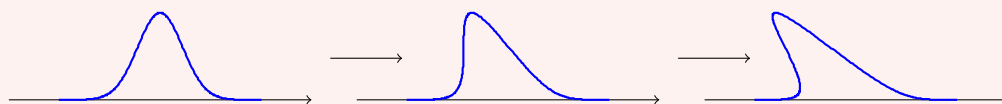
This is fine as long as the implicit relation can be solved for ξ . Let us see when it fails. Differentiating,

$$u_x = f'(\xi) \frac{\partial \xi}{\partial x}, \quad \frac{\partial \xi}{\partial x} = 1 - 6tf'(\xi) \frac{\partial \xi}{\partial x},$$

so

$$\frac{\partial \xi}{\partial x} = \frac{1}{1 + 6tf'(\xi)}, \quad u_x = \frac{f'(\xi)}{1 + 6tf'(\xi)}.$$

Whenever f' takes a negative value somewhere, this blows up at the finite time $t = -1/(6f'(\xi))$: the slope becomes infinite, and after that the implicit formula would give a multi-valued ‘solution’.



This is **wave breaking**, and it is what the nonlinear term $-6uu_x$ does on its own: taller parts of the wave move faster, overtake the parts in front of them, and the wave topples over.

So dispersion spreads a wave out and destroys it, and nonlinearity steepens a wave and destroys it. What happens if we include both?

§2.2 The Korteweg–de Vries Equation

Definition 2.3 (KdV equation)

The **Korteweg–de Vries equation**, or **KdV equation**, is

$$u_t + u_{xxx} - 6uu_x = 0.$$

The equation has a distinguished history. In 1834 the Scottish engineer John Scott Russell was watching a boat being drawn along a canal by a pair of horses. When the boat stopped suddenly, the water it had been pushing rolled forward as a single smooth heap, and Russell – to his eternal credit – got on his horse and chased it for a mile or

two along the canal, watching it travel without change of shape or speed. He called it the ‘wave of translation’. It took until 1895 for Korteweg and de Vries to write down the equation governing it.

Let us see, at least at the level of scaling, where the equation comes from.

Example 2.4 (Shallow water waves)

Consider small-amplitude surface waves on a layer of water of undisturbed depth h , and let $u(x, t)$ denote the surface elevation. Linear water wave theory gives the exact dispersion relation

$$\omega^2 = gk \tanh(kh).$$

For long waves, $kh \ll 1$, we expand $\tanh z = z - z^3/3 + O(z^5)$ to get

$$\omega^2 = gk \left(kh - \frac{(kh)^3}{3} + \dots \right) = c^2 k^2 \left(1 - \frac{(kh)^2}{3} + \dots \right), \quad c = \sqrt{gh},$$

so that for right-moving waves

$$\omega = ck \left(1 - \frac{k^2 h^2}{6} + \dots \right) = ck - \frac{ch^2}{6} k^3 + \dots.$$

A wave packet whose modes obey $\omega = ck - \beta k^3$ satisfies the linear PDE

$$u_t + cu_x + \beta u_{xxx} = 0, \quad \beta = \frac{ch^2}{6},$$

since $\partial_t \rightarrow -i\omega$ and $\partial_x \rightarrow ik$.

Now put back the leading nonlinearity. A finite-amplitude long wave rides on water of local depth $h + u$, so it travels a little faster where it is taller; carrying out the expansion properly gives a correction to the propagation speed of $\frac{3}{2}cu/h$, and hence a term γuu_x with $\gamma = 3c/2h$. Putting both effects together,

$$u_t + cu_x + \gamma uu_x + \beta u_{xxx} = 0.$$

Finally we normalise. Passing to a frame moving at speed c kills the cu_x term, and setting $u = Av(y, s)$ with $y = x - ct$ and $s = t/\beta$ gives

$$v_s + \frac{\gamma A}{\beta} vv_y + v_{yyy} = 0.$$

Choosing $A = -6\beta/\gamma$ produces

$$v_s + v_{yyy} - 6vv_y = 0,$$

which is KdV. So the equation is not some arbitrary combination of terms: it is what one gets from *any* weakly nonlinear, weakly dispersive, unidirectional wave system at leading order. This universality is why KdV turns up in plasma physics, in nonlinear optics, in blood flow and in lattice dynamics as well as in canals.

Some structural remarks are worth making before we solve anything. The KdV equation is invariant under

- (i) translations $x \mapsto x + a$, $t \mapsto t + b$;

- (ii) the scaling $u \mapsto \lambda^2 u$, $x \mapsto \lambda^{-1} x$, $t \mapsto \lambda^{-3} t$;
- (iii) the Galilean boost $u(x, t) \mapsto u(x - 6ct, t) + c$ for constant c .

The third is easy to miss and will matter later: adding a constant to u is the same as moving to a uniformly translating frame. The scaling symmetry tells us that there is really only a one-parameter family of solitary waves, all obtainable from one another by rescaling.

§2.3 Travelling Waves and the One-Soliton Solution

We now look for travelling wave solutions of KdV, and we shall find Russell's wave of translation. Let

$$u(x, t) = f(\xi), \quad \xi = x - ct,$$

for some speed c , and impose the boundary conditions $f, f', f'' \rightarrow 0$ as $|\xi| \rightarrow \infty$ (we want a localised wave on still water). Substituting,

$$-cf' + f''' - 6ff' = 0.$$

Every term is a derivative, so we can integrate once:

$$-cf + f'' - 3f^2 = A$$

for a constant A , and letting $\xi \rightarrow \infty$ forces $A = 0$. Now multiply by the integrating factor f' :

$$-cff' + f''f' - 3f^2f' = 0,$$

and integrate again, noting that $f''f' = \frac{1}{2} \frac{d}{d\xi} (f')^2$:

$$-\frac{c}{2}f^2 + \frac{1}{2}(f')^2 - f^3 = B,$$

and again $B = 0$ by the boundary conditions. Rearranging,

$$(f')^2 = 2f^3 + cf^2 = f^2(2f + c). \quad (\dagger)$$

This is a first-order ODE which we could grind through by separation of variables. It is quicker to guess. Since $(f')^2 \geq 0$ we need $2f + c \geq 0$ wherever $f \neq 0$; if we look for a solution with $f < 0$ (which is the natural sign for KdV in this normalisation) this forces $c > 0$ and $-c/2 \leq f \leq 0$. A function which decays at both ends and has a minimum of $-c/2$ suggests sech^2 , so we try

$$f(\xi) = -\frac{c}{2} \text{sech}^2(\alpha\xi)$$

and see what α must be. Using $\frac{d}{d\theta} \text{sech } \theta = -\text{sech } \theta \tanh \theta$,

$$f'(\xi) = c\alpha \text{sech}^2(\alpha\xi) \tanh(\alpha\xi), \quad (f')^2 = c^2\alpha^2 \text{sech}^4 \tanh^2.$$

On the other side of $(\dagger)$,

$$f^2(2f + c) = \frac{c^2}{4} \text{sech}^4 \cdot c(1 - \text{sech}^2) = \frac{c^3}{4} \text{sech}^4 \tanh^2,$$

using $1 - \text{sech}^2 = \tanh^2$. So $(\dagger)$ holds precisely when $c^2\alpha^2 = c^3/4$, that is $\alpha = \frac{1}{2}\sqrt{c}$. We have found our solution.

Proposition 2.5 (One-soliton solution)

For any $c > 0$, the function

$$u(x, t) = -\frac{c}{2} \operatorname{sech}^2 \left(\frac{\sqrt{c}}{2} (x - ct - x_0) \right)$$

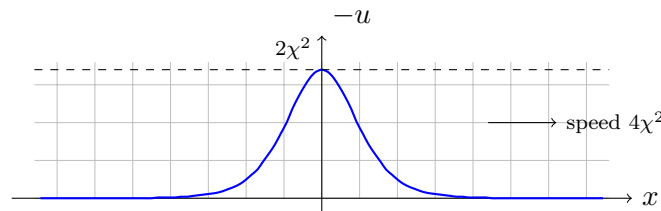
solves the KdV equation. Writing $c = 4\chi^2$ with $\chi > 0$, this takes the standard form

$$u(x, t) = -2\chi^2 \operatorname{sech}^2 (\chi(x - 4\chi^2 t - x_0)).$$

We call this a **soliton**. Look at what it says: the amplitude is $2\chi^2$, the speed is $4\chi^2$, and the width is $1/\chi$. So

taller solitons are faster and narrower.

This single fact is responsible for almost everything interesting that follows.



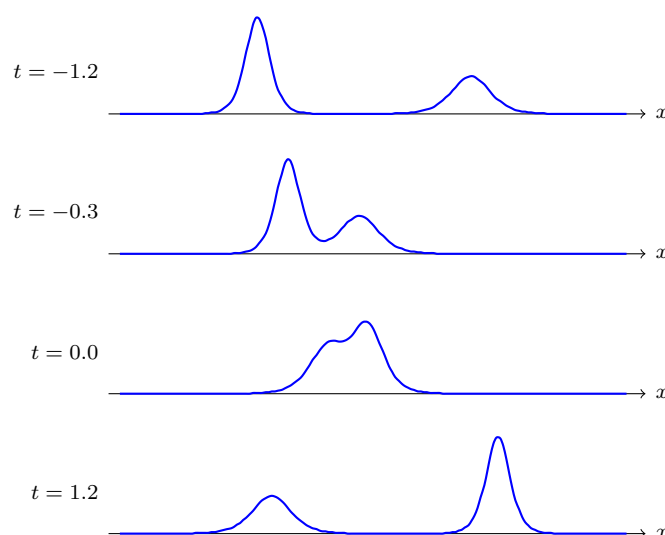
The balance is delicate. The dispersive term wants to spread the wave out; the nonlinear term wants to steepen its leading edge. For this particular profile, at this particular relation between height and speed, the two exactly cancel and the wave propagates unchanged forever.

Remark. The parameter χ is not a typographical accident. When we come to the inverse scattering transform we will find that $-\chi^2$ is precisely the eigenvalue of the Schrödinger operator $-\partial_x^2 + u$ associated with this potential. A one-soliton solution is the potential with exactly one bound state.

§2.4 Two Solitons, and Why Solitons Are Particles

Now for the interesting question. Suppose we start with two solitons, well separated, the taller one behind. Since the taller one is faster, it will catch up with the shorter one, and at some point they will overlap. The KdV equation is nonlinear, so we cannot simply add the two solutions; and we have every reason to expect that the collision will be messy, generating radiation and destroying both waves.

Here is what actually happens. (We will derive the exact formula in Section 3; for now we simply plot it, with $\chi_1 = 1$ and $\chi_2 = 1.6$. The window follows the interaction – both humps are in fact moving steadily to the right, and we plot $-u$ so that the solitons point upwards.)



The two solitons pass straight through one another and re-emerge with exactly their original shapes and speeds. Nothing is radiated away, nothing is lost. The only trace of the interaction is a **phase shift**: the taller soliton ends up slightly ahead of where it would have been, and the shorter one slightly behind. One can show that these shifts are

$$\Delta_2 = +\frac{1}{\chi_2} \log \frac{\chi_2 + \chi_1}{\chi_2 - \chi_1}, \quad \Delta_1 = -\frac{1}{\chi_1} \log \frac{\chi_2 + \chi_1}{\chi_2 - \chi_1},$$

for the taller and shorter solitons respectively, so the taller one gains and the shorter one loses.

This is extraordinary behaviour for a nonlinear PDE, and it is the reason for the name: a ‘soliton’ is a solitary wave which behaves like a *particle* in collisions. The phenomenon was discovered numerically by Zabusky and Kruskal in 1965, who were simulating KdV as a continuum model for the Fermi–Pasta–Ulam problem and were startled to find their pulses surviving collisions intact. It was their work that started the modern theory of integrable systems.

Note also that in the middle panel, at the moment of collision, the combined profile is not the sum of the two solitons, nor even taller than the larger one – the interaction is genuinely nonlinear. Superposition is violated at every instant; it is only the asymptotics that are clean.

§2.5 Conservation Laws

If solitons survive collisions, something must be preventing energy from leaking into other modes. The something is an infinite family of conservation laws. Let us find the first few by hand.

We will always assume that u and its derivatives decay rapidly as $|x| \rightarrow \infty$, so that we may integrate by parts freely and discard boundary terms.

Definition 2.6 (Conservation law)

A **conservation law** for an evolution equation is a relation of the form

$$\frac{\partial T}{\partial t} + \frac{\partial X}{\partial x} = 0,$$

where the **density** T and the **flux** X are functions of x, t, u and the derivatives of u . If such a relation holds for all solutions, then

$$\frac{d}{dt} \int_{-\infty}^{\infty} T \, dx = -[X]_{-\infty}^{\infty} = 0,$$

so $\int T \, dx$ is a first integral.

Proposition 2.7

The KdV equation possesses the conservation laws

$$\begin{aligned} \partial_t(u) + \partial_x(u_{xx} - 3u^2) &= 0, \\ \partial_t(u^2) + \partial_x(2uu_{xx} - u_x^2 - 4u^3) &= 0, \\ \partial_t(u^3 + \tfrac{1}{2}u_x^2) + \partial_x(\cdots) &= 0. \end{aligned}$$

Hence

$$I_1 = \int u \, dx, \quad I_2 = \int u^2 \, dx, \quad I_3 = \int (u^3 + \tfrac{1}{2}u_x^2) \, dx$$

are first integrals of KdV.

Proof. The first is just KdV rewritten: $u_t = -u_{xxx} + 6uu_x = -\partial_x(u_{xx} - 3u^2)$.

For the second, multiply KdV by $2u$:

$$(u^2)_t = 2uu_t = 2u(-u_{xxx} + 6uu_x) = -2uu_{xxx} + 12u^2u_x.$$

We must write the right-hand side as $-\partial_x X$. Observe that

$$\partial_x(2uu_{xx} - u_x^2 - 4u^3) = 2u_xu_{xx} + 2uu_{xxx} - 2u_xu_{xx} - 12u^2u_x = 2uu_{xxx} - 12u^2u_x,$$

which is exactly $-(u^2)_t$, as required.

For the third we argue slightly differently, since guessing the flux is tedious. Set $I_3 = \int (u^3 + \tfrac{1}{2}u_x^2) \, dx$ and differentiate:

$$\begin{aligned} \frac{dI_3}{dt} &= \int (3u^2u_t + u_xu_{xt}) \, dx \\ &= \int (3u^2 - u_{xx}) u_t \, dx, \end{aligned}$$

integrating the second term by parts. Now substitute $u_t = \partial_x(3u^2 - u_{xx})$ from the first conservation law, and write $w = 3u^2 - u_{xx}$. Then

$$\frac{dI_3}{dt} = \int w w_x \, dx = \frac{1}{2} \int \partial_x(w^2) \, dx = 0. \quad \square$$

Physically, I_1 is the mass, I_2 is the momentum, and I_3 will turn out to be the Hamiltonian

for KdV, which we may reasonably call the energy. The pattern continues: there is a conserved density involving u^4 , another involving u^5 , and so on forever. Finding them one at a time by inspection quickly becomes hopeless, and in Section 3 we will produce the whole infinite family in one stroke using the scattering problem.

Remark. Not every conservation law is interesting. For instance $\partial_t(u_x) + \partial_x(u_{xx} - 3u^2)_x = 0$ gives the first integral $\int u_x \, dx = 0$, which is true but vacuous, since it holds for any decaying function whatsoever. We will meet a systematic supply of such trivial laws later, and we will have to learn to discard them.

§2.6 The Miura Transform

We finish this section with a piece of magic whose significance will only become clear in Section 3. Alongside KdV sits its close relative.

Definition 2.8 (mKdV)

The **modified KdV equation** is

$$v_t + v_{xxx} - 6v^2v_x = 0.$$

The two equations look similar but are genuinely different (mKdV, for instance, is invariant under $v \mapsto -v$, which KdV is not). Nevertheless, they are related.

Proposition 2.9 (Miura transform)

Define the **Miura transform**

$$u = v^2 + v_x.$$

Then

$$u_t + u_{xxx} - 6uu_x = (2v + \partial_x)(v_t + v_{xxx} - 6v^2v_x).$$

In particular, if v solves mKdV then $u = v^2 + v_x$ solves KdV.

Proof. This is a direct computation. From $u = v^2 + v_x$ we get

$$\begin{aligned} u_t &= 2vv_t + v_{xt}, \\ u_x &= 2vv_x + v_{xx}, \\ u_{xxx} &= 6v_xv_{xx} + 2vv_{xxx} + v_{xxxx}, \\ 6uu_x &= 6(v^2 + v_x)(2vv_x + v_{xx}) = 12v^3v_x + 6v^2v_{xx} + 12vv_x^2 + 6v_xv_{xx}. \end{aligned}$$

Adding up,

$$\begin{aligned} u_t + u_{xxx} - 6uu_x &= 2vv_t + v_{xt} + 6v_xv_{xx} + 2vv_{xxx} + v_{xxxx} \\ &\quad - 12v^3v_x - 6v^2v_{xx} - 12vv_x^2 - 6v_xv_{xx} \\ &= 2v(v_t + v_{xxx} - 6v^2v_x) + (v_{xt} + v_{xxxx} - 6v^2v_{xx} - 12vv_x^2). \end{aligned}$$

On the other hand, writing $M = v_t + v_{xxx} - 6v^2v_x$,

$$\partial_x M = v_{xt} + v_{xxxx} - 6(2vv_x \cdot v_x + v^2v_{xx}) = v_{xt} + v_{xxxx} - 12vv_x^2 - 6v^2v_{xx},$$

which is exactly the bracketed expression above. So the two sides agree. $\square$

This is already useful – it lets us manufacture solutions of KdV from solutions of mKdV. But the reason it is a landmark result is the following observation. The relation

$$u = v^2 + v_x$$

is a **Riccati equation** for v , and the standard trick for Riccati equations is to substitute

$$v = \frac{\psi_x}{\psi}.$$

Then

$$v_x = \frac{\psi_{xx}}{\psi} - \frac{\psi_x^2}{\psi^2}, \quad v^2 = \frac{\psi_x^2}{\psi^2},$$

so $u = \psi_{xx}/\psi$, that is

$$-\psi_{xx} + u\psi = 0.$$

Better still, KdV has the Galilean symmetry noted above, so if u solves KdV then so does $u + \lambda$ suitably translated; applying the Miura transform to the shifted equation gives

$$-\psi_{xx} + u\psi = \lambda\psi.$$

Out of nowhere, the time-independent Schrödinger equation has appeared, with the solution of KdV playing the role of the potential. Gardner, Greene, Kruskal and Miura took this hint seriously in 1967, and what came out of it is the subject of the next section.